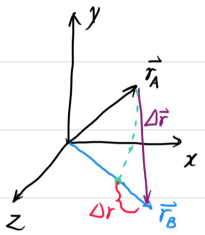
大物II力学

LL

质点运动



1. 位置矢量

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$\begin{cases} \cos\alpha = \dfrac{x}{r} \\ \cos\beta = \dfrac{y}{r} \\ \cos\gamma = \dfrac{z}{r} \end{cases}$$

位移

$\vec{x} = \vec{r}_B - \vec{r}_A$ $\quad (\Delta\vec{r})$ $\quad |\Delta\vec{r}| \neq \Delta r$

单方向直线 $\quad |\Delta\vec{r}| = \Delta s$, $\quad |d\vec{r}| = ds$

☆ $\quad |d\vec{r}| = ds$

平均速度：$\overline{\vec{v}} = \dfrac{\Delta\vec{r}}{\Delta t}$

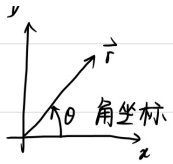
平均速率：$\overline{v} = \dfrac{\Delta s}{\Delta t}$

速度 $\quad \vec{v} = \dfrac{d\vec{r}}{dt}$ $\quad v = \dfrac{ds}{dt}$

平均速度 $\quad \overline{\vec{a}} = \dfrac{\Delta\vec{v}}{\Delta t}$

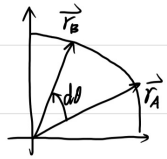
加速度 $\quad \vec{a} = \dfrac{d\vec{v}}{dt} = \dfrac{d^2\vec{r}}{dt^2}$

2. 圆周运动



$$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}$$

$d\vec{\theta}$ 右手则方向



平均角速度 $\bar{\omega} = \frac{\Delta\theta}{\Delta t}$

角速度 $\omega = \frac{d\theta}{dt}$ $\left(\vec{\omega} = \frac{d\vec{\theta}}{dt}\right)$

平均角加速度 $\bar{\beta} = \frac{\Delta\omega}{\Delta t}$

角加速度 $\beta = \frac{d\omega}{dt}$

$ds = \overset{\frown}{AB} = r d\theta$ $v = \omega r$

$d\vec{s} = d\vec{\theta} \times \vec{r}$

★ $\vec{v} = \vec{\omega} \times \vec{r}$

加速度 $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}(\vec{\omega} \times \vec{r}) = \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt}$

匀速率圆周 ω:定数

$$\vec{a} = \vec{\omega} \times \frac{d\vec{r}}{dt} = \vec{\omega} \times \vec{v} = \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

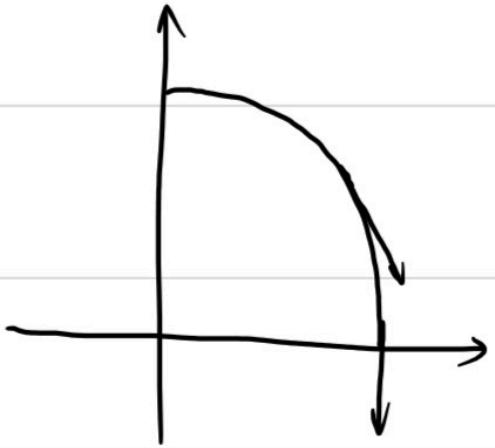
$$\vec{a} = -\omega^2 \vec{r}$$

★ $a = \omega^2 r = \frac{v^2}{r}$ 法向

变速率 $\frac{d\vec{\omega}}{dt} \times \vec{r} = \vec{\beta} \times \vec{r} = r\beta \hat{e}_\tau = \frac{dv}{dt}\vec{e}_\tau$

切向加速度

$$\begin{cases} ds = r\,d\theta . \\ \vec{a_\tau} = \dfrac{dv}{dt} \\ \vec{a_n} = \dfrac{v^2}{r} \Big/ \omega^2 r \end{cases}$$



$$a = \sqrt{a_\tau^2 + a_n^2}$$

$$\begin{cases} v = v_0 + at & \omega = \omega_0 + \beta t \\ x = v_0 t + \frac{1}{2} a t^2 + x_0 & \theta = \theta_0 + \omega_0 t + \frac{1}{2}\beta t^2 \end{cases}$$

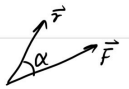
3. 质点动力

牛顿三定律：1. 惯 2. 动量 3. 相互作用

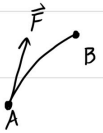
★ $\frac{d\vec{p}}{dt} = \vec{F} = \frac{d(m\vec{v})}{dt} = m\frac{d\vec{v}}{dt} = m\vec{a}$

功：力对空间的累积作用

$W = \vec{F}\cdot\vec{r} = Fr\cos\alpha$



$W = \sum \vec{F_i}\cdot\vec{r_i} = \int_A^B \vec{F}\cdot d\vec{r}$

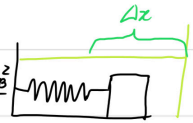


多力功：各个力功代数和

功率：平均功率：$\frac{\Delta W}{\Delta t}$

功率：$P = \vec{F}\cdot\vec{v}$

- 动能：$E_k = \frac{1}{2}mv^2$
- 引力功：$\int_A^B -\frac{GMm}{r^2}dr = \frac{GMm}{r_B} - \frac{GMm}{r_A}$
- 重力功：$\int_{h_A}^{h_B} -mg\,dz = mgh_A - mgh_B$
- 弹力功：$\int_{x_A}^{x_B} \vec{F}\cdot d\vec{x} = \int_{x_A}^{x_B} -kx\,dx = \frac{kx_A^2}{2} - \frac{kx_B^2}{2}$



- 势能：

$E_P = mgh$　$E_G = -\frac{GMm}{r}$　$E_e = \frac{1}{2}kx^2$

$W = \int_A^B \vec{F_{保}}\cdot d\vec{r} = E_{PA} - E_{PB}$

动量：力对时间。

$$\vec{F} = \frac{d\vec{p}}{dt} \qquad \vec{p} = m\vec{v}$$

冲量：$\vec{I} = \int_{t_1}^{t_2} \vec{F}\,dt$

动量定理：$\vec{I} = \Delta\vec{p}$

质点系：$\Delta\vec{p} = \vec{I}_{外合力}$

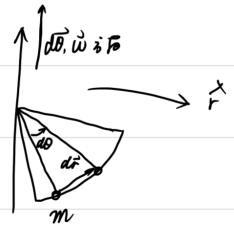
碰撞。

刚体.

$d\vec{r} = d\vec{\theta} \times \vec{r}$

$dr = r d\theta$

$\vec{a} = a_\tau \hat{\tau} - a_n \hat{r}$

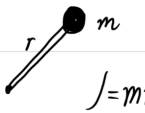


力矩

★ $\vec{M} = \vec{r} \times \vec{F}$

转动惯量.

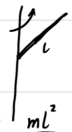
$J = \sum \Delta m_i r_i^2$



$J = mr^2$



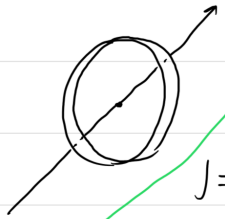
$J = \frac{ml^2}{12}$



$J = \frac{ml^2}{3}$



$J = \frac{mR^2}{2}$

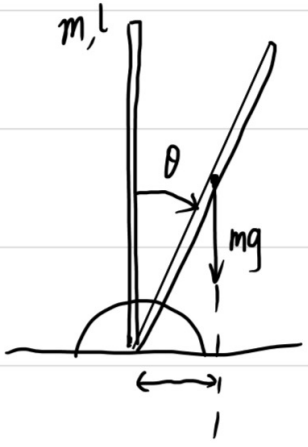


$J = mR^2$

$J_{球壳} = \frac{2}{5} mR^2$

$J_{球壳} = \frac{2}{3} mR^2$

$\vec{M} = J\vec{\beta} \quad \Leftrightarrow \quad \vec{F} = m\vec{a}$

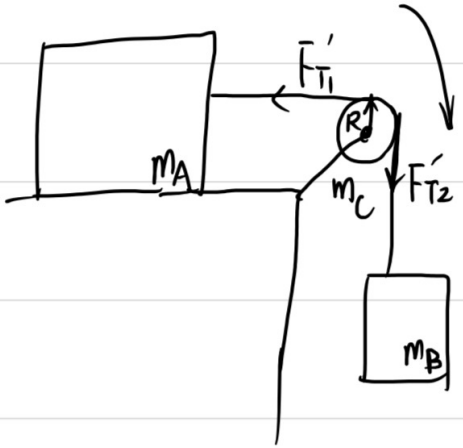


$$M = mg \cdot \frac{l\sin\theta}{2} = \frac{ml^2}{3}\beta \qquad \beta = \frac{3g\sin\theta}{2l}$$

$$\beta = \frac{d\omega}{dt} = \frac{d\omega}{d\theta}\frac{d\theta}{dt} = \omega\frac{d\omega}{d\theta}$$

$$\frac{3g}{2l}\int_0^\theta \sin\theta\, d\theta = \int_0^\omega \omega\, d\omega \quad ★$$

$$\omega = \sqrt{\frac{3g}{l}(1-\cos\theta)}$$



$$\begin{cases} m_B a_B = m_B g - \bar{F}_{T_2} \\ m_A a_A = \bar{F}_{T_1} \\ J\beta = \frac{m_C R^2}{2}\beta = F_{T_2}R - F_{T_1}R \\ a_A = a_B = R\beta = a \end{cases}$$

力矩功：

$$W = \int_{\theta_1}^{\theta_2} M \, d\theta$$

动能定理：$W = \frac{J}{2}\omega_2^2 - \frac{J}{2}\omega_1^2$

动能：$E_k = \frac{J\omega^2}{2}$

角动量：$\vec{L} = \vec{r} \times \vec{p} = m\vec{r} \times \vec{v}$

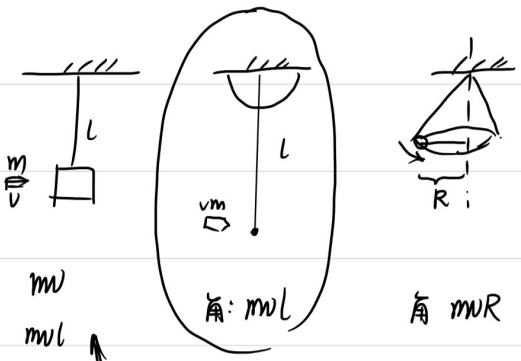
$L = mrv$

$\vec{M} = \frac{d\vec{L}}{dt}$

$\vec{L} = J\vec{\omega} = mr^2\vec{\omega}$

角动量定理：$\Delta \vec{L} = \int_{t_1}^{t_2} \vec{M} \, dt$

守恒：



水平：mv

角：mvl

角：mvl

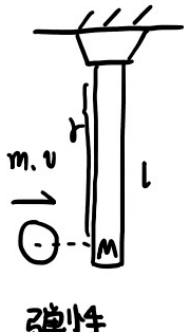
角 mvR

• -->



$$mvr = \frac{Ml^2}{3}\omega + mr^2\omega \qquad \omega = \frac{mvr}{\frac{Ml^2}{3} + mr^2}$$

$$\frac{1}{2}J\omega^2 = \left(\frac{Mgl}{2} + mgr\right)\text{versin}\varphi$$



碰撞

$$mvr = \frac{Ml^2}{3}\omega - mv'r$$

$$\frac{1}{2}mv^2 = \frac{1}{2}mv'^2 + \frac{Ml^2\omega^2}{6}$$

$$\frac{Ml^2\omega^2}{6} = \frac{Mgl}{2}\text{versin}\varphi$$